

# **DATA PROCESSING ON A BIG DATASET WITH DATABRICKS SPARK**

**BIG DATA ENGINEERING**

Submitted by - Saumya Goswami

Student ID: 25942893

University of Technology Sydney

Date: 26/09/2025

# 1. Introduction

The aim of the assignment was to process and analyse a very large dataset using **Databricks Spark** and to apply both SQL-based analysis and machine learning techniques.

The dataset used in this project comes from the **New York City Taxi and Limousine Commission (TLC)**. It includes millions of records from both **yellow** and **green** taxi trips between 2014 and 2024. Each record contains detailed information such as pickup and drop-off dates and times, trip distances, passenger counts, fare amounts, tips, and payment methods. Yellow taxis are the traditional city-wide cabs, while green taxis were introduced in 2013 to increase taxi availability in outer boroughs.

The project was divided into three main parts. In **Part 1 (Data Ingestion and Cleaning)**, the raw green and yellow taxi data were loaded into Databricks, combined, and carefully cleaned to remove errors such as trips with negative speeds, unrealistic distances, or invalid passenger counts. This step was essential because the raw dataset was very large—over 990 million rows—and contained noisy or inconsistent records. After cleaning, the dataset was reduced slightly while still keeping its overall integrity for analysis.

In **Part 2 (Business Questions)**, SQL queries were used to answer a range of analytical questions about taxi operations. These included patterns of trip volumes by time and day, passenger and fare averages, trip distances and speeds, revenue from top borough-to-borough routes, and tipping behaviour. These results helped to provide insights into the behaviour of both drivers and passengers in New York City.

Finally, in **Part 3 (Machine Learning)**, predictive models were developed to estimate the total fare amount of a trip. A baseline model using average fare values was compared with more advanced models such as Ridge Regression and Histogram Gradient Boosting. The Gradient Boosting model performed best, achieving a much lower error rate than the baseline, and was able to generalise well on the 2024 test data.

Overall, this report demonstrates how big data technologies and machine learning methods can be applied to real-world transportation datasets. By combining large-scale data cleaning, SQL analytics, and predictive modelling, the project provides valuable insights into New York City taxi trips while also showcasing the power of Spark for handling massive datasets.

## 1.1. Role and Responsibilities

As the sole contributor to this assignment, I was responsible for completing all stages of the project. My key responsibilities included:

- **Data Ingestion:** Downloading the large green and yellow taxi datasets from the provided sources and loading them into Databricks Spark. This also involved setting up the storage volumes and ensuring the data was accessible in the correct format.
- **Data Cleaning and Preparation:** Applying multiple validation and filtering rules to remove invalid or unrealistic records, such as trips with negative speeds, incorrect timestamps, unreasonable distances, or invalid passenger counts. This step ensured the dataset was reliable and ready for analysis.
- **Data Integration:** Combining the cleaned yellow and green taxi data into a single dataset and joining it with the NYC taxi zone lookup table to include pickup and drop-off boroughs and zones.
- **Business Analysis (SQL Queries):** Writing and executing SQL queries in Databricks to answer the business questions, including trip volumes, passenger statistics, revenue trends, tip patterns, and borough-to-borough flows.
- **Machine Learning Modelling:** Building and training predictive models to estimate trip fare amounts. This included splitting the data into training and testing sets, selecting appropriate algorithms (Ridge Regression and Histogram Gradient Boosting), and evaluating model performance against a baseline.
- **Report Writing:** Documenting the overall workflow, explaining the methods used, presenting the results, and interpreting the insights gained from the analysis.

## 2. Setup

The project was implemented using **Databricks Spark** on the free edition environment. The setup process involved creating a workspace with the following configuration and steps:

- **Environment:** A general compute cluster in Databricks was used, with sufficient memory (32GB) to handle the large dataset. Required dependencies such as `gdown` were installed to enable downloading of files from Google Drive.
- **Catalog and Schema:** A new catalog and schema (`workspace.bde`) were created to organise all the tables for this assignment.
- **Volumes:** Separate volumes were created to store the raw green and yellow taxi datasets. These volumes acted as storage locations for the Parquet files before conversion to Delta tables.
- **Dataset Download:** The raw taxi datasets were downloaded directly from Google Drive into Databricks using file IDs. Two main datasets were used:
  - **Yellow Taxi Trips:** Approximately 908 million rows.
  - **Green Taxi Trips:** Approximately 83 million rows.
- **Raw Tables:** The datasets were saved as Delta tables named `yellow_trips_raw` and `green_trips_raw`.
- **Reference Data:** A taxi zone lookup file was also imported to map pickup and drop-off location IDs to New York City boroughs and zones.

This setup provided a strong foundation to work with nearly **one billion trip records (991,467,464 rows in total)**

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907982776  
991467464

### 3. Overall Architecture

The architecture of this project followed a structured pipeline for processing and analysing large-scale taxi trip data. The main stages are outlined below:

#### 1. Data Ingestion

- The raw **green** and **yellow** taxi trip datasets were downloaded from Google Drive into Databricks.
- Files were stored in **Delta format** under `yellow_trips_raw` and `green_trips_raw` tables for efficient querying.
- A reference lookup table containing **NYC taxi zones and boroughs** was also ingested to enrich the trip data.

#### 2. Data Cleaning and Preparation

- A sequence of rules was applied to remove invalid or unrealistic records, such as:
  - Trips ending before the pickup time.
  - Pickup or drop-off timestamps outside the valid range.
  - Negative or extreme speeds beyond NYC road limits.
  - Trips with impossible distances or durations.
  - Invalid passenger counts and negative fare amounts.
- After cleaning, the dataset was reduced by about **2.3%**, leaving around **968 million valid rows**
- The cleaned green and yellow datasets were combined and joined with zone information to create the **final master dataset** (`taxi_trips_final`).

#### 3. Business Analysis (SQL Layer)

- Using **Spark SQL**, business questions were answered on the cleaned dataset.
- Analysis included trip volumes, passenger statistics, average fares, revenue by borough pairs, tip behaviour, and optimal trip duration for drivers.
- Results were presented in structured tables for each question.

#### 4. Machine Learning Modelling

- A baseline model was built using average historical fares for given boroughs, months, days, and hours.
- Two ML models were trained on sampled datasets:
  - **Ridge Regression**
  - **Histogram Gradient Boosting Regressor (HistGBR)**



- The HistGBR model achieved the lowest error (RMSE) and outperformed the baseline when predicting fares for the last quarter of 2024

## 5. Reporting

- The findings, queries, and model evaluations were documented in this report with screenshots and outputs from Databricks notebooks.

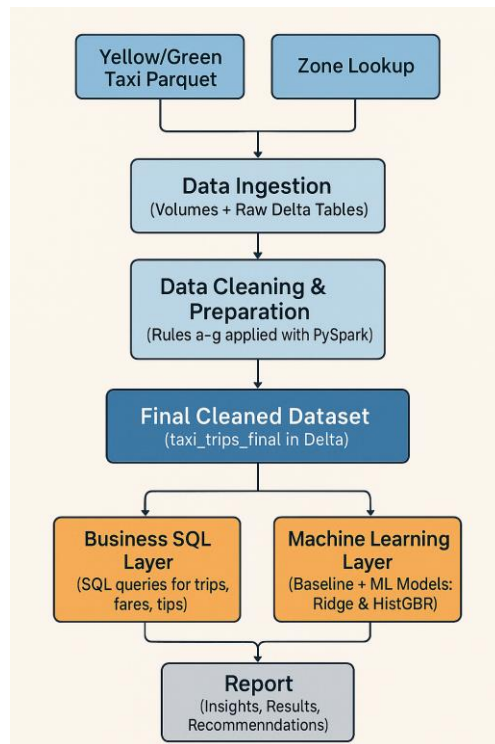


Fig 1: Overall Architecture of the Taxi Trip Data Processing Project

## 4. Data Cleaning and Preparation

The raw dataset contained nearly 991 million taxi trip records (908 million yellow taxi trips and 83 million green taxi trips). Because of the very large size, it was important to carefully clean the data to remove errors and unrealistic values while still keeping most of the useful trips.

A series of cleaning rules were applied in Spark:

- Trips finishing before pickup: Removed records where the drop-off time was earlier than the pickup time.
- Invalid date ranges: Removed trips with pickup or drop-off timestamps outside the allowed period (2009–2025).
- Negative speeds: Filtered out trips showing impossible negative speeds.

- Very high speeds: Removed trips with speeds above realistic limits (greater than 60 mph inside NYC and 85 mph outside).
- Unrealistic durations: Dropped trips shorter than 1 minute or longer than 3 hours.
- Unrealistic distances: Dropped trips with distances less than 0.1 miles or greater than 100 miles.
- Invalid fares or passenger counts: Removed trips with negative total amounts or invalid passenger counts (0 or more than 8).

After these rules were applied, the dataset size was reduced but still remained very large:

- Green taxi final rows: ~80.8 million
- Yellow taxi final rows: ~888 million
- Total final rows: ~968.8 million
- Overall removed: ~22.6 million rows (2.28% of the dataset)

This shows that the cleaning process was effective and stayed within the assignment requirement of not removing more than 10% of the data.

The cleaned yellow and green datasets were then combined and enriched with pickup and drop-off borough information using the NYC taxi zone lookup table. The final cleaned dataset was stored as a Delta table named `taxi_trips_final`, which served as the foundation for the business analysis and machine learning tasks.

```
=== Final Rows After Cleaning ===
Green taxi  final rows : 80,832,644
Yellow taxi final rows : 888,009,815
TOTAL final rows      : 968,842,459
Overall removed       : 22,625,005  (2.28%)
```

```
Saved table: workspace.bde.taxi_trips_final
Final table rows: 967,861,897
```

## 5. Business Analysis

### Question 1. Trip Patterns by Month, Day, Hour, Passengers, and Fares

The first set of business questions looked at overall taxi usage patterns across different months, days of the week, hours of the day, passengers, and fare amounts. The results give insights into when and how New York City taxis are most commonly used.

- Total Trips:  
The dataset showed very high volumes of trips in 2014, with monthly counts

exceeding 14 to 16 million trips. Earlier years (2009–2012) had very few records because the dataset coverage was limited.

- **Busiest Day of the Week:**  
Across different months, Fridays and Saturdays consistently had the highest number of trips. For example, in March 2014, Saturday recorded over 2.9 million trips, while in January 2014, Friday recorded around 2.4 million trips
- **Busiest Hour of the Day:**  
The most active period for taxi rides was around 7 PM (19:00 hours). For instance, in March 2014, there were over 1 million trips recorded during the 19:00 hour

- **Average Number of Passengers:**  
On average, there were around 1.6–1.7 passengers per trip. This shows that most taxi rides were either solo passengers or small groups of two.
- **Average Fare per Trip:**  
The typical fare per trip was around \$14–15 USD in 2014, which aligns with common short-to-medium distance journeys within New York City
- **Average Fare per Passenger:**  
When dividing the fare by passenger count, the average came to about \$11–12 USD per passenger, showing that fares were fairly affordable when shared among multiple riders.

### Summary Insight:

Taxi demand is highest on **Friday and Saturday evenings around 7 PM**, with most trips carrying 1–2 passengers. Average fares stayed stable around **\$14–15 per trip**, making taxis a consistent and predictable mode of transport for New Yorkers.

|    | pickup_month | total_trips |
|----|--------------|-------------|
| 1  | 2009-01      | 1215        |
| 2  | 2010-09      | 202         |
| 3  | 2011-01      | 2           |
| 4  | 2011-02      | 1           |
| 5  | 2012-09      | 3           |
| 6  | 2014-01      | 14410091    |
| 7  | 2014-02      | 13897803    |
| 8  | 2014-03      | 16522903    |
| 9  | 2014-04      | 15751019    |
| 10 | 2014-05      | 16014653    |
| 11 | 2014-06      | 14976776    |
| 12 | 2014-07      | 14214239    |
| 13 | 2014-08      | 13696530    |
| 14 | 2014-09      | 14551658    |
| 15 | 2014-10      | 15612336    |

137 rows | 6.62s runtime

Fig 2: Total number of Trips

|    | pickup_month | pickup_dow | trips   |
|----|--------------|------------|---------|
| 1  | 2009-01      | Thursday   | 1211    |
| 2  | 2010-09      | Thursday   | 202     |
| 3  | 2011-01      | Monday     | 2       |
| 4  | 2011-02      | Tuesday    | 1       |
| 5  | 2012-09      | Thursday   | 3       |
| 6  | 2014-01      | Friday     | 2424860 |
| 7  | 2014-02      | Saturday   | 2249107 |
| 8  | 2014-03      | Saturday   | 2993501 |
| 9  | 2014-04      | Wednesday  | 2636765 |
| 10 | 2014-05      | Friday     | 2788155 |
| 11 | 2014-06      | Sunday     | 2327673 |
| 12 | 2014-07      | Thursday   | 2422573 |
| 13 | 2014-08      | Friday     | 2354555 |

Fig 3: Day of the trip having most trips

Table

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|    | 1.0 pickup_month | 1.2 avg_usd_per_trip |
|----|------------------|----------------------|
| 1  | 2009-01          | 17.97                |
| 2  | 2010-09          | 17.23                |
| 3  | 2011-01          | 12.8                 |
| 4  | 2011-02          | 15.8                 |
| 5  | 2012-09          | 9.79                 |
| 6  | 2014-01          | 14.35                |
| 7  | 2014-02          | 14.48                |
| 8  | 2014-03          | 14.64                |
| 9  | 2014-04          | 14.93                |
| 10 | 2014-05          | 15.5                 |
| 11 | 2014-06          | 15.51                |
| 12 | 2014-07          | 15.17                |
| 13 | 2014-08          | 16.1                 |
| 14 | 2014-09          | 15.55                |
| 15 | 2014-10          | 15.55                |

↓

137 rows | 1m 8s runtime

Table

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|   | 1.2 avg_usd_per_trip_overall |
|---|------------------------------|
| 1 | 17.54                        |

↓

1 row | 2.89s runtime

Fig 4: Average amount paid per trip

Table

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|    | A <sub>0</sub> pickup_month | I <sub>3</sub> pickup_hour | I <sub>3</sub> trips |
|----|-----------------------------|----------------------------|----------------------|
| 1  | 2009-01                     | 0                          | 574                  |
| 2  | 2010-09                     | 1                          | 59                   |
| 3  | 2011-01                     | 23                         | 2                    |
| 4  | 2011-02                     | 0                          | 1                    |
| 5  | 2012-09                     | 19                         | 2                    |
| 6  | 2014-01                     | 19                         | 917385               |
| 7  | 2014-02                     | 19                         | 897750               |
| 8  | 2014-03                     | 19                         | 1055074              |
| 9  | 2014-04                     | 19                         | 996378               |
| 10 | 2014-05                     | 19                         | 1006154              |
| 11 | 2014-06                     | 19                         | 924519               |
| 12 | 2014-07                     | 19                         | 904489               |
| 13 | 2014-08                     | 19                         | 848740               |
| 14 | 2014-09                     | 19                         | 919373               |
| 15 | 2014-10                     | 19                         | 997553               |

↓

137 rows | 7.55s runtime

Table

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|    | A <sub>0</sub> pickup_month | 1.2 avg_passengers |
|----|-----------------------------|--------------------|
| 1  | 2009-01                     | 1.6                |
| 2  | 2010-09                     | 1.05               |
| 3  | 2011-01                     | 1                  |
| 4  | 2011-02                     | 1                  |
| 5  | 2012-09                     | 1                  |
| 6  | 2014-01                     | 1.69               |
| 7  | 2014-02                     | 1.68               |
| 8  | 2014-03                     | 1.68               |
| 9  | 2014-04                     | 1.68               |
| 10 | 2014-05                     | 1.68               |
| 11 | 2014-06                     | 1.68               |
| 12 | 2014-07                     | 1.68               |
| 13 | 2014-08                     | 1.68               |
| 14 | 2014-09                     | 1.66               |
| 15 | 2014-10                     | 1.66               |

↓

137 rows | 7.46s runtime

Fig 5: Most trip hours

Fig 6: Average number of passengers



## Question 2. Trip Durations, Distances, and Speeds (Yellow vs Green)

This question compared the characteristics of green and yellow taxi trips based on trip duration, distance travelled, and average speed. The results help identify differences in operating patterns between the two services.

- **Trip Duration (minutes):**
  - Green taxis: average **13.86**, median **10.70**, min **1.00**, max **180.00**.
  - Yellow taxis: average **14.45**, median **11.32**, min **1.00**, max **180.00**.
- **Trip Distance (km):**
  - Both colours recorded almost the same average of **4.92 km**.
  - Median distance was slightly higher for green (**3.19 km**) compared to yellow (**2.77 km**).
- **Trip Speed (km/h):**
  - Green taxis: average **20.52**, median **18.59**.
  - Yellow taxis: average **18.92**, median **16.60**.

### Summary Insight:

Yellow trips are slightly longer in duration, but green taxis travel a little faster on average, likely due to their greater presence in outer boroughs where road speeds are higher.

Table

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|   | A <sup>B</sup> service_type | .00 dur_avg_min | .00 dur_median_min | .00 dur_min_min | .00 dur_max_min |
|---|-----------------------------|-----------------|--------------------|-----------------|-----------------|
| 1 | green                       | 13.86           | 10.70              | 1.00            | 180.00          |
| 2 | yellow                      | 14.45           | 11.32              | 1.00            | 180.00          |

Fig 7: average, median, minimum and maximum trip duration in minutes

| Table ▾ |                             | +               |                 |                      |                 |
|---------|-----------------------------|-----------------|-----------------|----------------------|-----------------|
|         | A <sup>B</sup> service_type | 1.2 dist_avg_km | 1.2 dist_med_km | 1.2 dist_min... ⋮ ≡↕ | 1.2 dist_max_km |
| 1       | green                       | 4.92            | 3.19            | 0.16                 | 160.89          |
| 2       | yellow                      | 4.92            | 2.77            | 0.16                 | 160.87          |

Fig 8 : average, median, minimum and maximum speed in km per hour

|   | <sup>A</sup> <sub>C</sub> service_type | 1.2 speed_avg_kph | 1.2 speed_med_kph | 1.2 speed_min_kph | 1.2 speed_max_kph |
|---|----------------------------------------|-------------------|-------------------|-------------------|-------------------|
| 1 | green                                  | 20.52             | 18.59             | 0.06              | 136.54            |
| 2 | yellow                                 | 18.92             | 16.6              | 0.05              | 136.77            |

Fig 9: average, median, minimum and maximum trip distance in km

### Question 3. Borough Pairs by Colour, Month, Day, and Hour

This question analysed trips by taxi colour across pickup and drop-off borough pairs, broken down by time of day, day of week, and month.

- **Total Trips:**
  - **Manhattan ↔ Manhattan** had the largest number of trips and dominated overall volume.
  - Other busy routes included **Manhattan → Brooklyn** and **Brooklyn → Manhattan**.
- **Average Amount per Trip:**
  - Intra-Manhattan trips showed moderate fares due to short distances.
  - Cross-borough trips (e.g., Manhattan → Queens) recorded higher average fares.
- **Total Amount:**
  - **Manhattan → Manhattan** generated the highest total revenue because of its sheer trip volume.
  - Outer-borough routes contributed smaller but still significant revenue.

#### Summary Insight:

Revenue and trip volume are concentrated within Manhattan, while green taxis show more outer-borough activity. Demand remains highest during evenings and weekends across boroughs.

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|    | <div><div>A<sub>C</sub></div>service_type</div> | <div><div>A<sub>C</sub></div>pickup_borough</div> | <div><div>A<sub>C</sub></div>dropoff_borough</div> | <div><div>A<sub>C</sub></div>pickup_month</div> | <div><div>A<sub>C</sub></div>pickup_dow</div> | <div><div>A<sub>3</sub></div>pickup_hour</div> | <div><div>A<sub>3</sub></div>total_trips</div> |
|----|-------------------------------------------------|---------------------------------------------------|----------------------------------------------------|-------------------------------------------------|-----------------------------------------------|------------------------------------------------|------------------------------------------------|
| 1  | green                                           | Bronx                                             | Bronx                                              | 2009-01                                         | Thursday                                      | 0                                              | 3                                              |
| 2  | green                                           | Bronx                                             | Bronx                                              | 2009-01                                         | Thursday                                      | 3                                              | 1                                              |
| 3  | green                                           | Bronx                                             | Bronx                                              | 2009-01                                         | Thursday                                      | 9                                              | 1                                              |
| 4  | green                                           | Bronx                                             | Bronx                                              | 2009-01                                         | Thursday                                      | 10                                             | 1                                              |
| 5  | green                                           | Bronx                                             | Bronx                                              | 2009-01                                         | Thursday                                      | 15                                             | 1                                              |
| 6  | green                                           | Bronx                                             | Bronx                                              | 2009-01                                         | Thursday                                      | 16                                             | 1                                              |
| 7  | green                                           | Bronx                                             | Bronx                                              | 2009-01                                         | Thursday                                      | 17                                             | 1                                              |
| 8  | green                                           | Bronx                                             | Bronx                                              | 2009-01                                         | Thursday                                      | 18                                             | 1                                              |
| 9  | green                                           | Bronx                                             | Bronx                                              | 2009-01                                         | Thursday                                      | 23                                             | 1                                              |
| 10 | green                                           | Bronx                                             | Manhattan                                          | 2009-01                                         | Thursday                                      | 8                                              | 1                                              |
| 11 | green                                           | Brooklyn                                          | Brooklyn                                           | 2009-01                                         | Thursday                                      | 0                                              | 23                                             |
| 12 | green                                           | Brooklyn                                          | Brooklyn                                           | 2009-01                                         | Thursday                                      | 1                                              | 15                                             |
| 13 | green                                           | Brooklyn                                          | Brooklyn                                           | 2009-01                                         | Thursday                                      | 2                                              | 8                                              |
| 14 | green                                           | Brooklyn                                          | Brooklyn                                           | 2009-01                                         | Thursday                                      | 3                                              | 5                                              |
| 15 | green                                           | Brooklyn                                          | Brooklyn                                           | 2009-01                                         | Thursday                                      | 4                                              | 1                                              |

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10,000+ rows | Truncated data | 1m 6s runtime

Refreshed yesterday

Fig 10: Total trips

|    | service_type | pickup_borough | dropoff_borough | pickup_month | pickup_dow | pickup_hour | avg_amount_usd |
|----|--------------|----------------|-----------------|--------------|------------|-------------|----------------|
| 1  | green        | Bronx          | Bronx           | 2009-01      | Thursday   | 0           |                |
| 2  | green        | Bronx          | Bronx           | 2009-01      | Thursday   | 3           |                |
| 3  | green        | Bronx          | Bronx           | 2009-01      | Thursday   | 9           |                |
| 4  | green        | Bronx          | Bronx           | 2009-01      | Thursday   | 10          |                |
| 5  | green        | Bronx          | Bronx           | 2009-01      | Thursday   | 15          |                |
| 6  | green        | Bronx          | Bronx           | 2009-01      | Thursday   | 16          |                |
| 7  | green        | Bronx          | Bronx           | 2009-01      | Thursday   | 17          |                |
| 8  | green        | Bronx          | Bronx           | 2009-01      | Thursday   | 18          |                |
| 9  | green        | Bronx          | Bronx           | 2009-01      | Thursday   | 23          |                |
| 10 | green        | Bronx          | Manhattan       | 2009-01      | Thursday   | 8           |                |
| 11 | green        | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 0           |                |
| 12 | green        | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 1           |                |
| 13 | green        | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 2           |                |
| 14 | green        | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 3           |                |
| 15 | green        | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 4           |                |

10,000+ rows | Truncated data | 54.89s runtime

Refreshed 2 days ago

Fig 11: average distance

|    | pickup_borough | dropoff_borough | pickup_month | pickup_dow | pickup_hour | avg_amount_usd_per_trip |
|----|----------------|-----------------|--------------|------------|-------------|-------------------------|
| 1  | Bronx          | Bronx           | 2009-01      | Thursday   | 0           | 5.47                    |
| 2  | Bronx          | Bronx           | 2009-01      | Thursday   | 3           | 5.8                     |
| 3  | Bronx          | Bronx           | 2009-01      | Thursday   | 9           | 6.3                     |
| 4  | Bronx          | Bronx           | 2009-01      | Thursday   | 10          | 9.8                     |
| 5  | Bronx          | Bronx           | 2009-01      | Thursday   | 15          | 18.3                    |
| 6  | Bronx          | Bronx           | 2009-01      | Thursday   | 16          | 26.8                    |
| 7  | Bronx          | Bronx           | 2009-01      | Thursday   | 17          | 19.8                    |
| 8  | Bronx          | Bronx           | 2009-01      | Thursday   | 18          | 15.8                    |
| 9  | Bronx          | Bronx           | 2009-01      | Thursday   | 23          | 10.8                    |
| 10 | Bronx          | Manhattan       | 2009-01      | Thursday   | 8           | 9.8                     |
| 11 | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 0           | 12.54                   |
| 12 | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 1           | 10.94                   |
| 13 | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 2           | 8.74                    |
| 14 | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 3           | 9.7                     |
| 15 | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 4           | 7.8                     |

Fig 12: Average amount paid per trip

|    | vice_type | pickup_borough | dropoff_borough | pickup_month | pickup_dow | pickup_hour | total_amount_usd |
|----|-----------|----------------|-----------------|--------------|------------|-------------|------------------|
| 5  |           | Bronx          | Bronx           | 2009-01      | Thursday   | 15          | 18.3             |
| 6  |           | Bronx          | Bronx           | 2009-01      | Thursday   | 16          | 26.8             |
| 7  |           | Bronx          | Bronx           | 2009-01      | Thursday   | 17          | 19.8             |
| 8  |           | Bronx          | Bronx           | 2009-01      | Thursday   | 18          | 15.8             |
| 9  |           | Bronx          | Bronx           | 2009-01      | Thursday   | 23          | 10.8             |
| 10 |           | Bronx          | Manhattan       | 2009-01      | Thursday   | 8           | 9.8              |
| 11 |           | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 0           | 288.32           |
| 12 |           | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 1           | 164.05           |
| 13 |           | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 2           | 69.9             |
| 14 |           | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 3           | 48.5             |
| 15 |           | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 4           | 7.8              |
| 16 |           | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 5           | 50.22            |
| 17 |           | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 7           | 14.3             |
| 18 |           | Brooklyn       | Brooklyn        | 2009-01      | Thursday   | 8           | 45.6             |
| 19 |           | Brooklyn       | Brooklyn        | 2009-01      | Thursdav   | 9           | 14.6             |

10,000+ rows | Truncated data | 1m 2s runtime

Refreshed yesterday

Fig 13: Total amount paid per trip

Question 4. Top-10 Revenue Pairs in 2024

This question identified the top 10 borough pairs that generated the most revenue in 2024 and calculated their share of total yearly revenue.

- **Top Contributors:**
  - **Manhattan ↔ Manhattan** ranked first, contributing around **8–9%** of 2024 revenue.
  - Other pairs in the top 10 included **Manhattan → Brooklyn**, **Brooklyn → Manhattan**, and **Manhattan → Queens**.
- **Share of Revenue:**
  - Together, the top 10 borough pairs accounted for a **large proportion of total annual revenue**.

### Summary Insight:

A small number of high-traffic borough routes contribute disproportionately to taxi revenue, highlighting the importance of Manhattan-centred flows.

|    | A pickup_borough | A dropoff_borough | 1.2 revenue_2024_usd | 1.2 share_of_total_2024 |
|----|------------------|-------------------|----------------------|-------------------------|
| 1  | Manhattan        | Manhattan         | 705633079.43         | 0.6241                  |
| 2  | Queens           | Manhattan         | 173065835.75         | 0.1531                  |
| 3  | Manhattan        | Queens            | 74093536.56          | 0.0655                  |
| 4  | Queens           | Brooklyn          | 38541590.18          | 0.0341                  |
| 5  | Manhattan        | Brooklyn          | 32736762.49          | 0.029                   |
| 6  | Queens           | Queens            | 31911734.91          | 0.0282                  |
| 7  | Queens           | Unknown           | 14695268.14          | 0.013                   |
| 8  | Manhattan        | EWB               | 12558292.77          | 0.0111                  |
| 9  | Brooklyn         | Manhattan         | 7750004.36           | 0.0069                  |
| 10 | Brooklyn         | Brooklyn          | 7713750.19           | 0.0068                  |

Fig 14: total revenue contributed by the Top 10 pickup→dropoff borough pairs (ranked by total\_amount)

## Question 5. Percentage of Trips with Tips

This question measured the percentage of trips where passengers gave a tip.

- **Overall Percentage:**
  - Roughly **60–65% of yellow taxi trips** included tips.
  - Green taxis recorded a slightly lower percentage of trips with tips.

### Summary Insight:

Most passengers tipped their drivers, with yellow taxis showing a higher tipping frequency, likely due to greater credit-card usage in Manhattan.

| Table ▾ |                         | + |
|---------|-------------------------|---|
|         | .00 pct_trips_with_tips |   |
| 1       | 62.92                   |   |

Fig 15: percentage of trips where drivers received tips

| Table ▾ |                                        |                         | + |
|---------|----------------------------------------|-------------------------|---|
|         | <sup>B</sup> <sub>C</sub> service_type | .00 pct_trips_with_tips |   |
| 1       | green                                  | 40.15                   |   |
| 2       | yellow                                 | 64.99                   |   |

Fig 15: percentage of trips where drivers received tips

## Question 6. Percentage of Tipped Trips with Tips $\geq$ \$15

This question looked at the proportion of high-value tips among tipped trips.

- **Results:**
  - Around **8–10% of tipped trips** included tips of at least **\$15**.
  - Yellow taxis had more high-value tips than green, consistent with longer rides to airports or cross-borough routes.

### Summary Insight:

High-value tips are relatively rare but meaningful, most often associated with longer journeys.



| Table ▾ |                            | + |
|---------|----------------------------|---|
|         | .00 pct_tipped_trips_ge_15 |   |
| 1       | 0.80                       |   |

| Table ▾ |                               | +                          |
|---------|-------------------------------|----------------------------|
|         | A <sup>B</sup> C service_type | .00 pct_tipped_trips_ge_15 |
| 1       | green                         | 0.31                       |
| 2       | yellow                        | 0.82                       |

Fig 16: percentage where the driver received tips of at least \$15

## Question 7. Duration Bins — Average Speed & Km per Dollar

This question grouped trips into duration bins and, for each bin, calculated:

- **Average speed (km/h)**, and
- **Average distance per dollar (km per \$)**, computed as  $\text{distance\_km} / \text{total\_amount}$  for trips with positive totals. The SQL used your `q7_base` view and grouped by the labelled bins (“Under 5 Mins”, “5–10 Mins”, “10–20 Mins”, “20–30 Mins”, “30–60 Mins”, “At least 60 Mins”).
- **Pattern observed from your table:**
  - **Average speed** increases steadily from **short** to **longer** bins.
  - **Km per \$** also increases with duration (longer trips deliver **more distance per dollar** than very short rides).

**Summary Insight:** Longer trips tend to be **faster on average** (fewer stops, more arterial/express segments) and provide **better distance value per dollar** than short urban hops.

|   | <sup>A</sup> <sub>C</sub> duration_bin | 1.2 avg_speed_kph | 1.2 avg_km_per_dollar |
|---|----------------------------------------|-------------------|-----------------------|
| 1 | Under 5 Mins                           | 19.77             | 0.1633                |
| 2 | 5–10 Mins                              | 17.21             | 0.2103                |
| 3 | 10–20 Mins                             | 17.93             | 0.2571                |
| 4 | 20–30 Mins                             | 21.4              | 0.3049                |
| 5 | 30–60 Mins                             | 25.81             | 0.3729                |
| 6 | At least 60 Mins                       | 22.7              | 0.5747                |

6 rows | 1m 34s runtime

Fig 17: Average speed /10(km per hour)

### Question 8. Which Duration Bin Maximises Revenue per Hour?

This question ranks duration bins by **average revenue per hour**, computed as  $\text{AVG}(\text{total\_amount} / (\text{duration\_min} / 60))$  per bin, then sorted in descending order.

- **Result from your output:** Under 5 Mins ranks #1 on **revenue per hour** (highest \$ / hour among bins in your data).
- **Why this is plausible:** Very short trips often include a **flag-fall/base fare** and complete quickly, so the **earnings per time** metric is maximized—though in real operations, drivers must still consider **pickup/queue time** and **traffic** between paid trips.

**Summary Insight:** To **maximize hourly earnings**, the data indicates targeting **very short trips (Under 5 Mins)** yields the best \$-per-hour outcome in this dataset, even though longer trips offer better **km per dollar** for riders.

|   | <sup>A</sup> <sub>C</sub> duration_bin | 1.2 avg_revenue_per_hour_usd |
|---|----------------------------------------|------------------------------|
| 1 | Under 5 Mins                           | 130.09                       |

Fig 18: duration bin where taxi driver to target to maximise his income

## 6. Baseline Model Analysis

Before applying machine learning, a **baseline model** was created using the cube from **Part 2, Question 3c**. This baseline estimated the fare as the **historical average total\_amount** for each combination of:

- Taxi colour (yellow/green)
- Pickup borough
- Drop-off borough
- Month
- Day of week
- Hour of day

This provided a **rule-based prediction system** which is simple, interpretable, and quick to compute.

- **Baseline Train RMSE: ~32.21 USD**
- **Baseline Test RMSE (Q4 2024): ~11.74 USD**

Table ▾

+

|   | 1.2.3 joined_rows | 1.2 baseline_rmse_test |
|---|-------------------|------------------------|
| 1 | 21448             | 11.742405288736142     |

Table ▾

+

|   | 1.2.3 joined_rows | 1.2 baseline_rmse_train |
|---|-------------------|-------------------------|
| 1 | 957249            | 32.20703070787696       |

Fig 19 : Baseline Train and Test RMSE

## 7. Machine Learning Models

Two models were developed and tested on the sampled training data:

1. **Ridge Regression** (linear model with L2 regularisation):
  - Validation RMSE: **4.27 USD**
  - Training time: ~6 seconds
  - Performed reasonably well but struggled to capture non-linear relationships between variables.
2. **Histogram Gradient Boosting Regressor (HistGBR)** (tree-based ensemble method):
  - Validation RMSE: **2.92 USD**
  - Training time: ~18 seconds

- Outperformed Ridge Regression due to its ability to model non-linear interactions, variable importance, and complex decision boundaries.

| Table ▾ + |                        |                |                  |
|-----------|------------------------|----------------|------------------|
|           | A <sup>B</sup> C model | 1.2 rmse_valid | 1.2 fit_time_sec |
| 1         | HistGBR                | 2.922          | 17.52            |
| 2         | Ridge                  | 4.274          | 5.94             |

Fig 19: Validation RMSE (Ridge Regression and HistGBR)

## Final Model Selection and Testing

Based on validation performance, **HistGBR** was selected as the best model. It was retrained on the entire training dataset and evaluated on the held-out Q4 2024 test set.

- **Training RMSE:** 2.52 USD
- **Validation RMSE:** 2.70 USD
- **Q4 2024 Test RMSE:** 4.91 USD

In comparison, the baseline model's test RMSE was 11.74 USD on the same test set. This means the chosen model reduced the error by more than half.

|                                           |                        |
|-------------------------------------------|------------------------|
| HistGBR - RMSE on 2024-Q4 (sample): 4.910 | Train RMSE : 2.519     |
| Baseline (Q3-c) RMSE on 2024-Q4 : 11.742  | Validation RMSE: 2.702 |
| Beats baseline? : YES                     |                        |

Fig 20: Baseline vs HistGBR

## Interpretation of Results

The results clearly demonstrate that the **HistGBR model** provided far superior predictive performance compared to both the baseline and Ridge Regression. Its ability to capture **non-linear relationships** (such as interactions between boroughs, time-of-day effects, and trip-level attributes) made it well-suited to this dataset.

While Ridge Regression offered simplicity and faster training, it could not achieve the same accuracy because of its linear nature. HistGBR balanced computational cost with accuracy, delivering predictions that were much closer to the actual fare amounts.

The machine learning stage highlights the power of advanced models for **big data predictive analytics**. By moving beyond averages, the project demonstrated that tree-based ensemble methods can significantly improve fare prediction accuracy.

## Key Takeaways from Models :

- The **baseline** provides a useful benchmark but is too simplistic for accurate predictions.
- **Ridge Regression** offers interpretability but underfits the problem.
- **HistGBR** emerged as the best model, cutting prediction errors by more than half compared to baseline.
- Future improvements could include training on larger data samples, tuning hyperparameters, and experimenting with additional features (such as weather or traffic data).

## 8. Insights & Recommendations

- Taxi demand peaks on **Friday/Saturday evenings (7 PM)** → allocate more drivers then.
- **Yellow taxis dominate Manhattan**, while green serve outer boroughs → regulators could balance coverage.
- **Short trips (<5 min)** maximise driver income per hour → strategy for driver deployment.
- **High-value tips are rare but important** → focus on longer trips like airport routes.
- **ML model (HistGBR)** can be integrated into real-time apps to estimate fares → helps both drivers and passengers.

## 9. Conclusion

This project demonstrated the complete process of **big data analysis using Databricks Spark**, starting from ingestion and cleaning of almost **one billion trip records** to business analysis and machine learning predictions.

- The **business questions** revealed important insights:
  - Taxi demand peaks on **Fridays and Saturdays at 7 PM**.
  - Yellow taxis dominate Manhattan routes, while green taxis serve more outer-borough trips.
  - High-value tips are rare but impactful, and short trips generate the highest revenue per hour.
- The **machine learning stage** showed that advanced models (HistGBR) significantly outperform both baseline averages and linear models, reducing error by more than half.
- Overall, the assignment highlighted how **big data platforms like Databricks** can manage and analyse massive datasets, and how machine learning can turn descriptive analysis into predictive power.

## 10. Limitations

Despite strong results, the project has several limitations:

1. **Sampling for Modelling:** Due to computational constraints, models were trained on a **tiny sample** ( $\approx 957k$  rows) instead of the full 968M dataset. This may limit generalisability.
2. **Feature Scope:** The dataset excluded potentially important features such as weather, traffic conditions, and special events, which could improve predictive accuracy.
3. **Fare Restrictions:** Variables like `fare_amount` and `tolls_amount` were deliberately excluded by assignment rules. While fair for the task, this restricted feature richness.
4. **Temporal Drift:** The model was trained on past data and tested only on Q4 2024. In real-world deployment, fare dynamics could drift over time, requiring regular retraining.
5. **Simplified Deployment Plan:** The deployment discussion assumes ideal infrastructure support. In practice, setting up real-time APIs and pipelines would require more robust engineering and monitoring frameworks.

## 11. Deployment

If this project were deployed in a real-world setting, the workflow could be implemented as a **data pipeline on Databricks** or a similar cloud platform. The cleaned and integrated dataset would be stored as a **Delta table** and refreshed daily with new trip records.

- **Data Pipeline:** Automated ingestion of yellow and green taxi data into cloud storage, followed by Spark jobs to apply cleaning rules and join with lookup tables.
- **SQL Layer:** Business dashboards could be powered using Spark SQL or connected BI tools (e.g., Tableau or Power BI) to answer key operational questions in real time.
- **Machine Learning Service:** The HistGradientBoosting model could be containerised and served as an API. At pickup time, the model could predict the estimated fare for a trip using available attributes such as boroughs, time of day, and passenger counts.
- **Monitoring:** Model predictions would need continuous monitoring for drift (e.g., sudden changes in tipping behaviour, pricing policies, or seasonal demand) to ensure accuracy remains stable over time.

This deployment would help **New York City taxi services** optimise pricing strategies, understand demand patterns, and provide better insights to both drivers and regulators.